When we apply a method such as Implicit Midpoint to a nonlinear molecular dynamics problem, we may reduce the equations to determining midpoint values $\hat{\boldsymbol{q}}$, $\hat{\boldsymbol{p}}$ such that

$$\hat{\boldsymbol{q}} = \boldsymbol{q}_n + \frac{h}{2}\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}, \qquad \hat{\boldsymbol{p}} = \boldsymbol{p}_n - \frac{h}{2}\nabla U(\hat{\boldsymbol{q}}),$$

where, then,

$$\boldsymbol{q}_{n+1} = \hat{\boldsymbol{q}} + \frac{h}{2}\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}, \qquad \boldsymbol{p}_{n+1} = \hat{\boldsymbol{p}} - \frac{h}{2}\nabla U(\hat{\boldsymbol{q}}).$$

We may therefore view the step as a composition of Implicit Euler and Explicit Euler half-steps. Eliminating $\hat{p}$, we find

$$\hat{\boldsymbol{q}} = \boldsymbol{q}_n + \frac{h}{2}\boldsymbol{M}^{-1}\boldsymbol{p}_n - \frac{h^2}{4}\boldsymbol{M}^{-1}\nabla U(\hat{\boldsymbol{q}}).$$

This is a nonlinear system of equations for $\hat{\boldsymbol{q}}$. There is no avoiding the substantial challenge of solving this system. In principle this can be achieved using various iterative methods, such as the Gauss-Newton method. In case the forces are long-ranged (e.g. due to Coulombic terms) the cost of a single iteration will include the cost of a force evaluation, although we must also cope with numerically solving a system of linear equations for the update. In order to benefit from the symplectic property of the Implicit Midpoint method, the iteration must be performed to high accuracy (i.e. several iterations will typically be needed), otherwise one will observe growing errors in energy and trajectory accuracy. Given the accuracy issue associated to using large stepsizes due to the corruption of the phase, the possible gain from using an implicit method is found to be limited and methods like this are rarely encountered in molecular modelling applications. An exception are the methods used in handling constrained systems (see Section 4.3, below).

4.2 Multiple Timestepping

The stringency of the timestep threshold and the failure of implicit methods to address the issue have led to alternative proposals for ameliorating the instability in molecular dynamics, in particular the method of "multiple timestepping" which we next describe. Variants of this method were described in a series of papers in the early 1990s by Tuckerman and Berne [374] and Tuckerman et al. [375, 377]. A simultaneous development by Grubmüller et al. [158] most closely matches the presentation given here. The first resonance analysis is due to Biesiadecki and Skeel [35].